

single2diff2single: single-ended ↔ differential analog pad buffers

Designer Christoph Maier **Reviewer** Tim Edwards **Date** 2026-10-09, 17:00 CEST

Repos [sg13cmos5l](#) · [gf180](#)

Analog-on-top macros that let fully differential on-chip blocks talk to single-ended pins: a single-ended → differential input buffer, a differential → single-ended class-AB output driver (gain $\frac{1}{2}$, $v_{\text{out}} - v_{\text{ref}} = (v_{\text{inp}} - v_{\text{inn}})/2$), their current-mode bias, pad enables and CDM protection. The IHP sg13cmos5l design is the reference. The GF180MCU version is a port, started 2026-10-06.

Links point at what is pushed to GitHub: sg13cmos5l branch pcells (865a7bb) and sude1_buecher (fad5834); gf180 branch main (deddafd) and sude1_buecher (a12ad6b). The GF180 schematic drawings below were exported from Christoph's local working tree, which includes the hand edits of 2026-10-09 (local commit 64f5edb, not yet pushed). Line numbers in the log links refer to the pushed versions.

Status at a glance

Chipalooza #2 — IHP

ihp-sg13cmos5l · top cell
sg13cmos5l_cm_ip__single2diff2single

Schematics Class-AB driver (MP-DDA + Miller), four bias variants, power-down cells: drafted, round-tripped (0 mismatches), still in _sudelbuecher/design_considerati
Layout **partial** Clamp PyCells reproduce IHP flat cells; IOPadDiff2Single clamp placement in KLayout; driver size mockups only. Top level still the template inverter.

Verification Template flow (DRC/LVS/PEX/CACE) proven on template cells. Driver CACE: first run 2026-10-02, 11 fails (baseline Miller design), superseded by the MP-DDA proposal.

Blocking Moving the driver from notes into macros/; slot/pin assignment; layout.

Chipalooza #3 — GF180MCU

gf180mcuD · top cell
gf180_cm_ip__single2diff2single
(tentative)

Schematics Six macros in xschem: ClassABDriver, ClassABBias (In/Out/bandgap BG/BGdn), OgueyAebischerBias (03v3, 06v0), PadEnable, both pads.

Layout **none** No layout exists; DRC/LVS/PEX targets not run.

Verification CACE datasheets for 11 cells, run in a cloud container (xschem 3.4.8RC, ngspice-42/47). Not yet reproduced in IIC-OSIC-TOOLS.

Blocking Harness repo still empty; slot size, analog-pin count and PR-boundary layer are guesses.

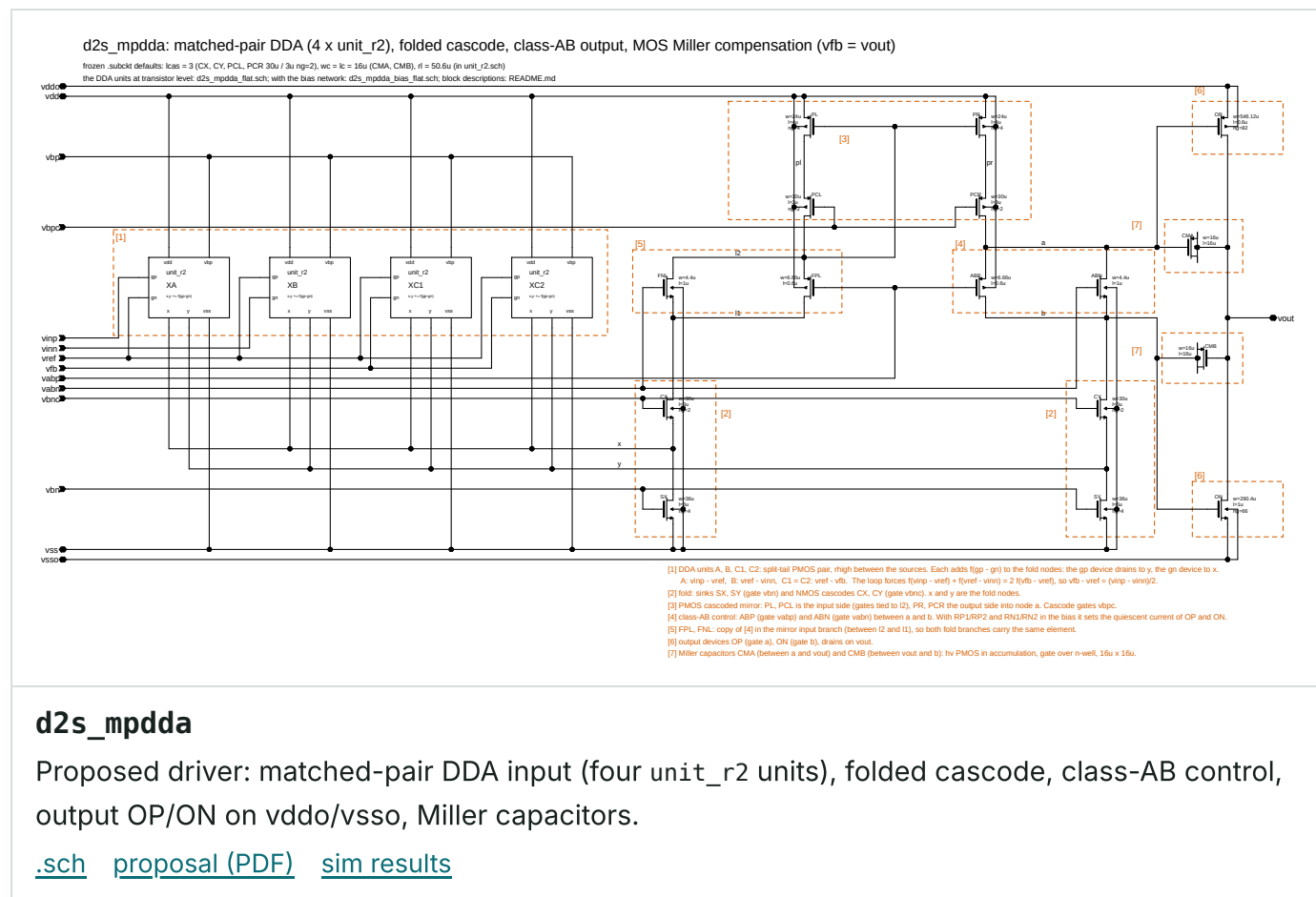
Questions for Tim

- Analog pins in a tiny slot.** The precheck inherited from the template rejects analog-pins > 0 unless slot-size is small, yet a tiny analog floorplan template ships. Which rule holds for sg13cmos5l and for GF180?
[sg13cmos5l README](#), [warning under submission.yaml](#)
- Slot and pin assignment in sg13cmos5l_oed_chipalooza.** Which slot number, and how do vin, vout and the shared vcm / vdiffp / vdiffn map onto sN_an[x]?
[pin_mapping_proposal_DRAFT.md](#) · [s2d_d2s_pinbuffers.md](#)

3. **Output-stage rail vsso:** vssio or vss_3v3? The driver keeps its output devices on separate rails vddo / vsso.
[improvements/xschem/README.md, "Output-stage rails"](#)
4. **Enable signal.** Working assumption: en_3v3 (0...3.3 V, active high, last port), power_3v3_ena static. Is a 1.2 V digital enable with a level shifter expected instead, and does the harness provide per-slot power switching?
[power_down/log.md L41-60 \(decisions\)](#)
5. **Case (a): the output devices double as the pad's ESD clamps** (OP / ON drawn as IHP clamp frames). Acceptable, or should the clamps stay separate? Unpowered behaviour: ON conducts 0.13–0.36 mA briefly into vsso while the floating rail charges through OP.
[power_down/log.md L120-170 \(results\)](#)
6. **GF180 analog pad.** gf180mcu_ocd_io__asig_5p0 has HBM diodes only; CDM protection is ours (Tim's > 50 Ω / > 25 μm). With CDM diodes to a 3.3 V vdd the input is not 5 V tolerant. Are 5 V levels on these pads a requirement?
[docs/OPERATING_LIMITS.md](#) · [esd_protection/README.md](#)
7. **GF180 harness interface.** gf180mcu_ocd_chipalooza was empty as of 2026-09-30; harness_stub/ holds our assumed interface. Slot size, analog-pin count and PR-boundary layer 0 are labelled guesses.
[harness_stub/](#) · [submission.yaml](#)
8. **Load range for the driver spec.** Phase margin over the load sweep drops to 24.9° (GF180) / 26.5° (IHP baseline). Which C_{load} / R_{load} range should the pad driver guarantee?
9. **Untrimmed bandgap.** GF180 ClassABBiasBG: ± 1.7 % over MOS corners $\times V \times T$, but poly-resistor spread gives -23 / +38 %. Acceptable for a bias reference, or add a trim?

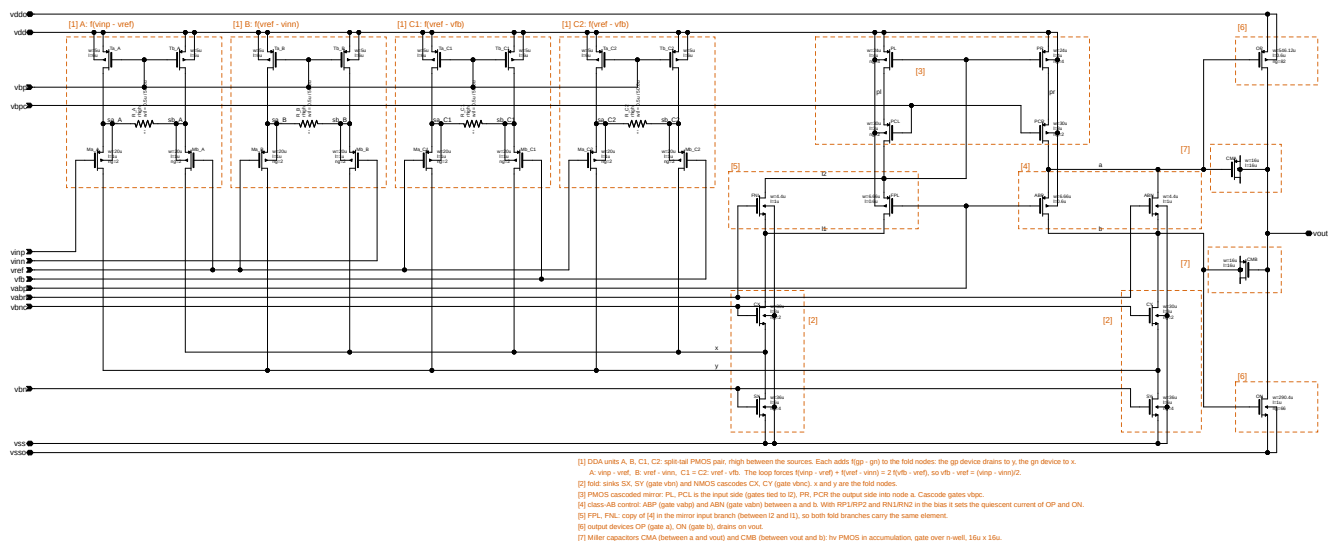
sg13cmos5l · schematics

The class-AB driver work lives in the notes worktree until it moves into macros/. The sheets were generated from SPICE decks, hand edited, and checked by netlisting back (check_xschem.py, MISMATCHES 0). The figures are exported from those sheets.



d2s_mpdda_flat: d2s_mpdda with the four DDA units (unit_r2) drawn at transistor level

same parts and devices as d2s_mpdda, unit devices and nets carry the unit name: Td_A, Td_B, R_A, R_B, Ma_A, Mb_A, Mh_A, Mh_B, ... rhigh body = vss
 frozen_subblock defaults: lcas = 3, wc = 1e, n = 50.6u, block descriptions: README.md



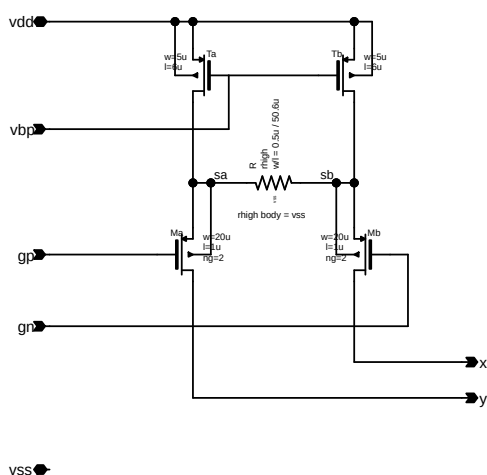
d2s_mpdda_flat

Same driver with the four DDA units at transistor level; blocks [1]–[7] tagged.

[.sch](#)

unit_r2: split-tail PMOS pair, rhigh degeneration (unit of d2s_mpdda)

gp drains to y, gn drains to x; x and y go to the folding nodes

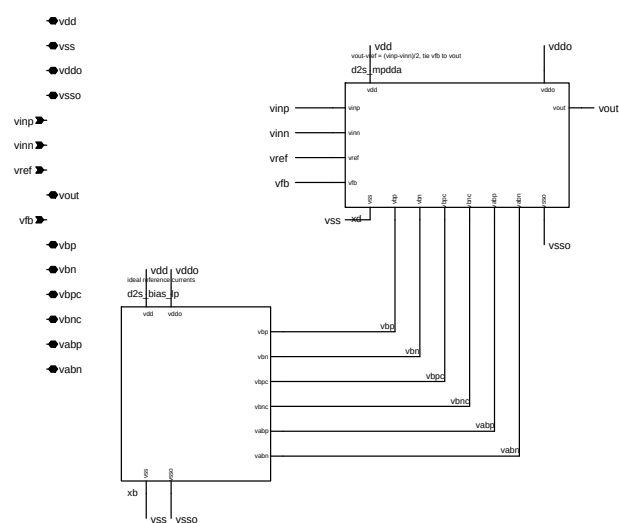


unit_r2

One DDA unit, $2 \times 5 \mu\text{A}$, resistor degenerated.

[.sch](#)

d2s_mpdda + d2s_bias_lp test fixture (CACE DUT); bias nets are pins for bias-quality perturbation



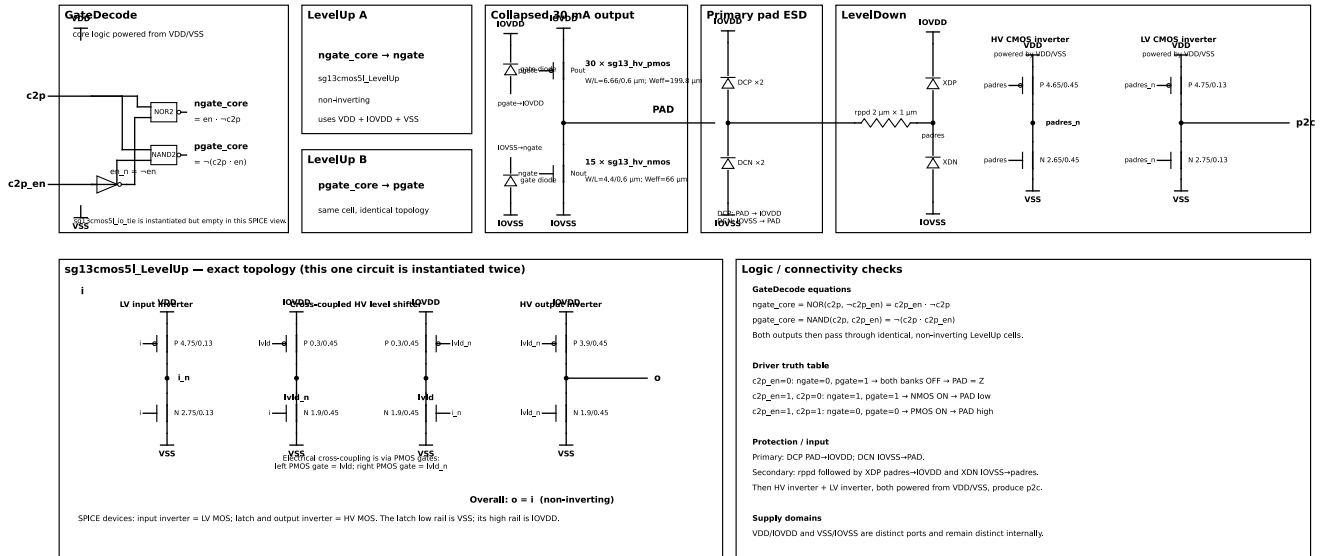
d2s_mpdda_biased

CACE fixture: driver + d2s_bias_lp, bias nets as pins.

[.sch](#)

sg13cmos5l_IOPadInOut30mA — reconstructed from official IHP SPICE

Important: VDD, VSS, IOVDD and IOVSS are separate nets. There is no internal short between either supply pair.



Authoritative source: IHP-GmbH/IHP-sg13cmos5l • lib.refsg13cmos5l_io/spice/sg13cmos5l_io.spi • drawing collapses only the parallel output MOS arrays; guard rings are omitted.

sg13cmos5l_IOPadInOut30mA (IHP, reconstructed)

Reference for pad-cell internals and clamp frames, drawn from the extracted hierarchy.

[hierarchy.spi](#) [pad_cell_internals.md](#)

Power-down variants (2026-10-06)

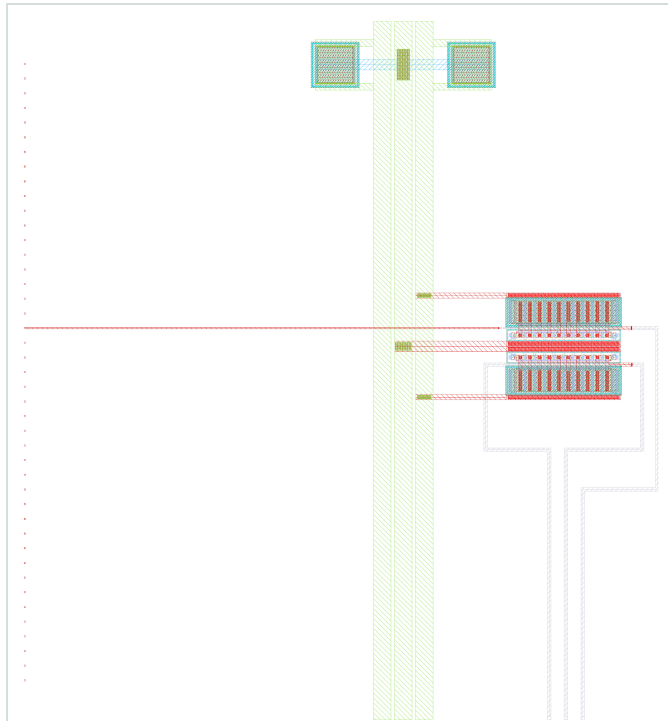
Five `_pd` cells and four testbenches, new switches in [E] frames, round trip MISMATCHES 0. Disabled supply current 0.08–0.35 nA at −40/27 °C and ≤ 47 nA at ff/3.6 V/125 °C, set by the clamp-frame output stage.

- [power_down/xschem/](#) — `d2s_mpdda_pd`, `d2s_bias_{in,out,oa,bg}_pd`, `tb_pd_{enable,leak,startup_oa,unpowered}`
- [results_power_down.txt](#)

Design-repo schematics (branch pcells)

- [sg13cmos5l_cm_ip_single2diff2single.sch](#) (top)
- [user_project_wrapper_3a.sch](#)
- [sg13cmos5l_IOPadDiff2Single.sch](#)
- [sg13cmos5l_IOPadSingle2Diff.sch](#)
- [OgneyAebischerBias.sch](#) · [ToBiasStartup.sch](#)
- [ClampN15N15.sch](#) · [ClampP15N15.sch](#)
- [DCNdiode.sch](#) · [DCPdiode.sch](#)
- [slot.sch](#)

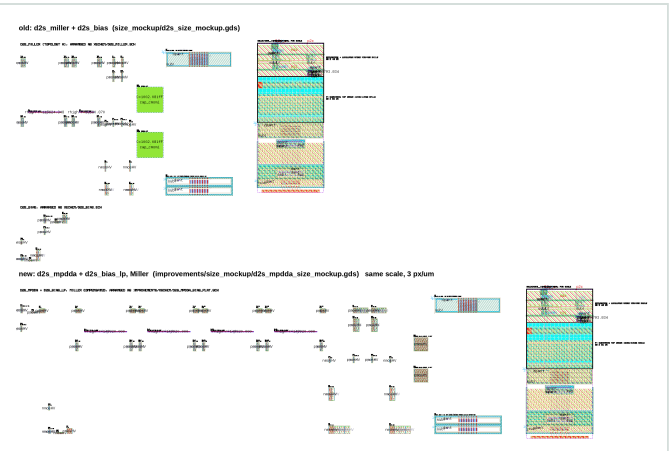
sg13cmos5l · layout



Top level, 200 × 200 μm (tiny)

Current render. Still instantiates the template's inverter cells.

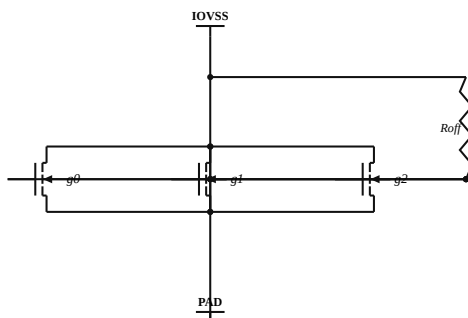
[.klay.gds](#) [final GDS](#)



d2s size mockup, old vs new

Area estimate of the Miller-compensated baseline against d2s_mpd2s (2026-10-03).

[generator.py](#) [.gds](#)



Clamp_N20N0D unit cell

Decomposition of IHP's flat clamp into a repeatable finger cell.

[DRC script](#)

Layout files

CELL / FILE	STATE	NOTES
scripts/pcells/ Clamp_N / Clamp_P	matches IHP	KLayout PyCells in the PDK's own style; parameter ng; identical to IHP's flat cells at the reference ng. N43N43D4R not covered.
IOPadDiff2Single/layout/klayout/	in progress	Clamp-cell substitution: manual PCell, scripted use_clamp_cell() swap, clamp size sweep, test.
macros/IOPad/layout/klayout/	reference	IHP pad cells: IOPadAnalog, InOut30mA, IOVdd/IOVss, Vdd/Vss.
macros/slot/layout/	scaffold	Slot macro with its own Makefile; KLayout LVS was being debugged (2026-09-28).
Class-AB driver, bias, enables	none	Size mockups only.

sg13cmos5l · verification

Baseline Miller driver, first CACE run (2026-10-02)

Schematic netlist, d2s_miller_biased. This is the design the MP-DDA proposal replaces; the fails below drove the redesign.

PARAMETER	LIMIT	MIN	TYP	MAX	STATUS
Phase margin	$\geq 45^\circ$	71.4°	74.8°	77.2°	pass
Phase margin vs load	$\geq 45^\circ$	26.5°	61.5°	86.8°	fail
PSRR at 1 kHz	$\geq 40\text{ dB}$	72.5 dB	72.5 dB	72.5 dB	pass
THD (h2..h7)	$\leq -40\text{ dB}$	-51.1 dB	-33.2 dB	-14.8 dB	fail
Output offset, mismatch	$\pm 10\text{ mV}$	-32.4 mV	-0.24 mV	48.7 mV	fail

PARAMETER	LIMIT	MIN	TYP	MAX	STATUS
Gain, process MC	0.485–0.515	0.415	0.511	0.605	fail
Settling to 1 %, rising	≤ 1000 ns	306 ns	955 ns	2533 ns	fail
Supply current	≤ 600 μ A	406 μ A	426 μ A	450 μ A	pass
Short- circuit current, source / sink	≥ 5 mA	59.8 / 74.2 mA	70.3 / 82.5 mA	80.4 / 92.2 mA	pass

- [first_run_summary.md](#) (all 50 rows) · [CACE datasheet with plots](#) · [d2s_miller_biased.yaml](#)
- [proposed_improvements.md](#) — the MP-DDA redesign with its own numbers; [run_improvements.py](#) reproduces them in one run (3 min 49 s).
- Template flow on the inherited cells: Magic and KLayout DRC clean, Magic+Netgen LVS "Circuits match uniquely", full-RC PEX, inverter CACE ([logs/i_claude](#), [logs/inverter_pex](#), [logs/counter_digital](#)).
- Oguey–Aebischer bias CACE: [OgueyAebischer.yaml](#) (generated datasheets not yet committed).

GF180 · schematics

All sheets exported from xschem with the gf180mcu_fd_pr symbols. Hierarchy:
gf180mcu_IOPadDiff2Single → BiasRefEnable → ClassABBiasIn → ClassABDriver (units ClassABUnitR) + DriverEnable / EnableInv. The bandgap ClassABBiasBG is not yet wired into the pad, which still takes iref from a pin.

gf180mcu_IOPadDiff2Single: class-AB differential -> single-ended pad driver with enable (PROOF OF CONCEPT, 2026-10-08)

Blocks: ClassABBiasIn (macros/ClassABBiasIn): bias tree and the six diodes, fed by iref (5 uA into the pad) through BiasRefEnable.
ClassABDriver (macros/ClassABDriver): matched pair DDA, folded cascode, class-AB output OP / ON, Miller compensation; vfb = out, so out = (inp - inv)/2.
DriverEnable, EnableInv (macros/PadEnable): en = 0 -> OP / ON gates tied to vddo / vsso, vabp / vabn to vdd / vss, iref cut off and N1's gate line to vss.
Bias enters as a current (iref), never as a voltage. vddo / vsso: output-stage rails (OP, ON and their replicas RP1, RN1).
Not included yet: CDM / ESD network at out (OP and ON are meant to double as the pad clamps, I4P case (a)); a level shifter for a 1.2 V enable.

The schematic diagram illustrates the internal structure of the `gf180mcu_IOPadDiff2Single` component. It is a differential Class-AB driver with an enable function. The circuit is composed of several key blocks:
1. **ClassABBiasIn**: A bias tree and six diodes, fed by `iref` (5 uA) through **BiasRefEnable**.
2. **ClassABDriver**: A matched pair DDA, folded cascode, class-AB output OP / ON, Miller compensation; `vfb = out`, so `out = (inp - inv)/2`.
3. **DriverEnable** and **EnableInv**: Control logic where `en = 0` -> OP / ON gates tied to `vddo / vsso`, `vabp / vabn` to `vdd / vss`, `iref` cut off and N1's gate line to `vss`.
4. **Bias**: Enters as a current (`iref`), never as a voltage. `vddo / vsso` are output-stage rails (OP, ON and their replicas RP1, RN1).
5. **CDM / ESD**: Not included yet. `out` (OP and ON) are meant to double as the pad clamps, I4P case (a); a level shifter for a 1.2 V enable.
The diagram shows various power rails (`vdd`, `vss`, `vddo`, `vsso`) and control signals (`inp`, `inn`, `vref`, `out`, `iref`, `en`, `en_b`). The output is labeled `out`.

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gf180mcu_IOPadDiff2Single.sch

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gf180mcu_IOPadDiff2Single

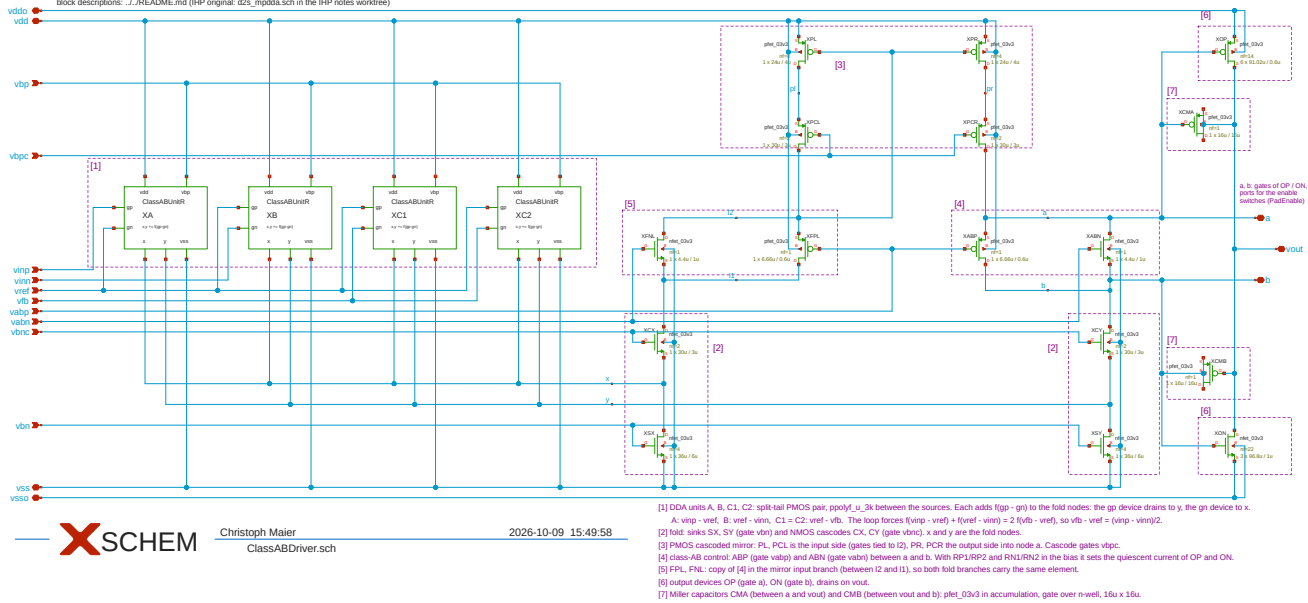
Output pad, block level. Ports `vdd` `vss` `vddo` `vsso` `inp` `inn` `vref` `out` `iref` `en`. CDM/ESD at `out` and a 1.2 V enable level shifter not yet included.

[.sch](#) [README](#)

GF180MCU (gf180mcuD) 03v3 port of the IHP sheet d2s_mpdda.sch, 2026-10-08. Wires and placement unchanged;
 devices plet_03v3 / nlet_03v3 / ppolyl_u_3k, sizes re-derived for GF180 (README.md).

d2s_mpdda: matched-pair DDA (4 x unit_r2), folded cascode, class-AB output, MOS Miller compensation (vfb = vout)

fixed sizes: lcas = 3 (CX, CY, PCL, PCR, 30u / 3u ng=2), wc = lc = 10u (CMA, CMB), R = ppolyl_u_3k 1u x 50u (in ClassABUnitR.sch)
 block descriptions: .../README.md (IHP original: d2s_mpdda.sch in the IHP notes worktree)



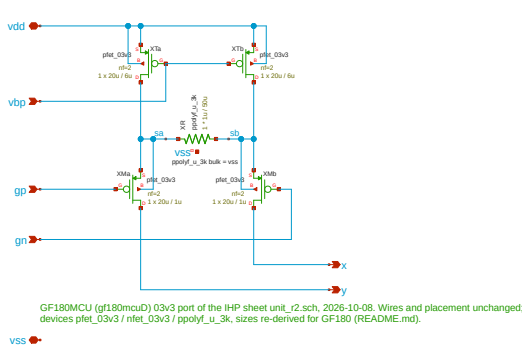
ClassABDriver

Driver core: matched-pair DDA, folded cascode, class-AB output OP/ON, Miller compensation;
 ports a, b for the enable.

[.sch](#) [README](#) [6 testbenches](#)

unit_r2: split-tail PMOS pair, ppolyl_u_3k degeneration (unit of d2s_mpdda)

gp drains to y, gn drains to x, x and y go to the folding nodes

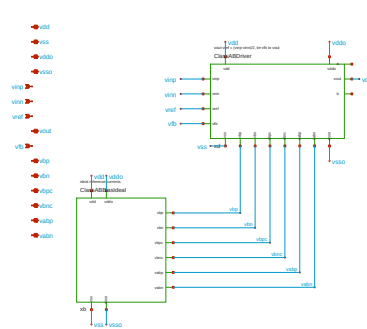


ClassABUnitR

One degenerated DDA unit.

[.sch](#)

d2s_mpdda + d2s_bias_1p test fixture (CACE DUT); bias nets are pins for bias-quality perturbation



ClassABDriverBiased

CACE fixture: driver + ideal bias.

[.sch](#)

ClassABiasBG

Self-contained current-mode (Banba) bandgap reference, 5 μA out of iout into the ClassABiasIn tree. GF180 re-design (2026-10-09) of the IHP bg_core:

[C] input branch PB / PBC $\rightarrow$ NB (gate line g) $\rightarrow$ eb: R1B to vss and R0 + Q2 (8 x pnp_05p00x05p00); diode branch PA / PAC $\rightarrow$ NA (diode, g) $\rightarrow$ ea: R1A and Q1.

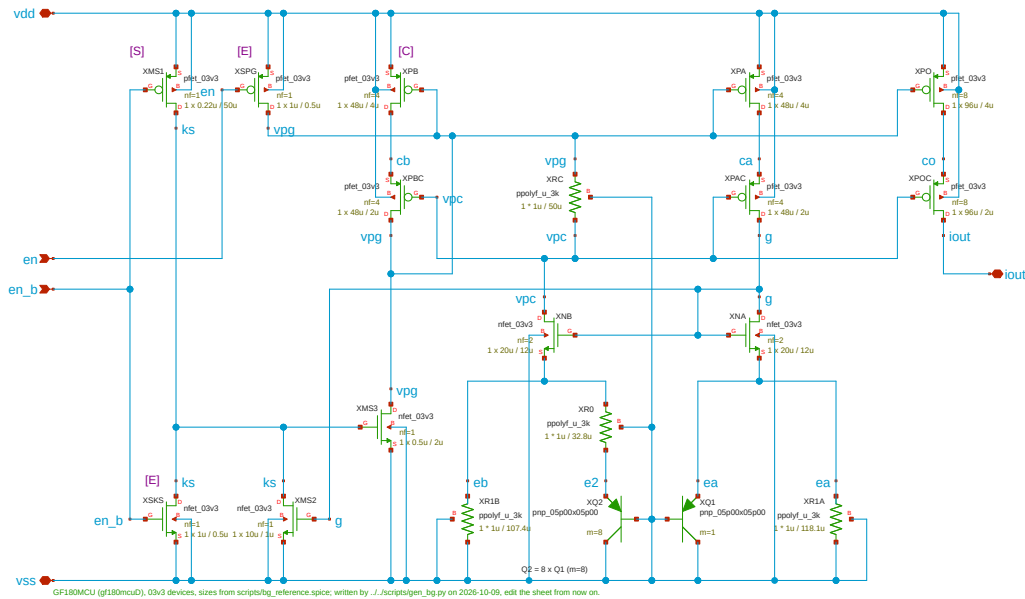
NA / NB force ea = eb, the mirror equal currents, so each branch carries $V_{BE1} / R1 + U_T \ln 8 / R0$ (32 % PTAT with ppolyf_u_3k's -0.15 %/K); PO / POC $\rightarrow$ 2 x $\rightarrow$ 5 μA .

Wide-swing cascode: the input branch's diode connection closes around PBC (vpg = its drain); the cascode gate line vpc = vpg - I RC (RC 150k), NB's drain on vpc.

R1A is 10 % longer than R1B: with both PNPs off the loop gain is > 1 , so the core cannot rest in the resistor-only state.

[S] start-up: MS1 weak pull-up (gate en_b) on ks, MS3 pulls vpg down while ks is high, MS2 releases ks once g is up. [E] RefCoreEnable: vpg $\rightarrow$ vdd, ks $\rightarrow$ vss while en = 0.

NA / NB bulks on vss (common substrate). The deep-n-well variant is ClassABiasBGdn.



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ClassABiasBG.sch

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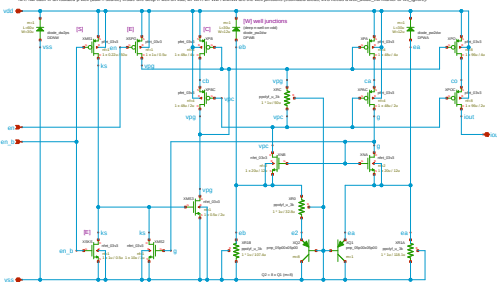
ClassABiasBG

Current-mode (Banba) bandgap, 5 μA out of iout. GF180 re-design: wide-swing cascode mirror, start-up on vpg, NMOS pair 20/12, ppolyf_u_3k, PNP ratio 8. Hand edited 2026-10-09 (enable switches SPG, SKS inline).

[.sch](#) [README](#), "ClassABiasBG"

ClassABiasBGdn

Self-contained current-mode (Banba) bandgap reference, 5 μA out of iout into the ClassABiasIn tree. GF180 re-design (2026-10-09) of the IHP bg_core:
[C] input branch PB / PBC $\rightarrow$ NB (gate line g) $\rightarrow$ eb: R1B to vss and R0 + Q2 (8 x pnp_05p00x05p00); diode branch PA / PAC $\rightarrow$ NA (diode, g) $\rightarrow$ ea: R1A and Q1.
NA / NB force ea = eb, the mirror equal currents, so each branch carries $V_{BE1} / R1 + U_T \ln 8 / R0$ (32 % PTAT with ppolyf_u_3k's -0.15 %/K); PO / POC $\rightarrow$ 2 x $\rightarrow$ 5 μA .
Wide-swing cascode: the input branch's diode connection closes around PBC (vpg = its drain); the cascode gate line vpc = vpg - I RC (RC 150k), NB's drain on vpc.
R1A is 10 % longer than R1B: with both PNPs off the loop gain is > 1 , so the core cannot rest in the resistor-only state.
[S] start-up: MS1 weak pull-up (gate en_b) on ks, MS3 pulls vpg down while ks is high, MS2 releases ks once g is up. [E] RefCoreEnable: vpg $\rightarrow$ vdd, ks $\rightarrow$ vss while en = 0.
NA / NB bulks on vss (common substrate). The deep-n-well variant is ClassABiasBGdn.



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ClassABiasBGdn.sch

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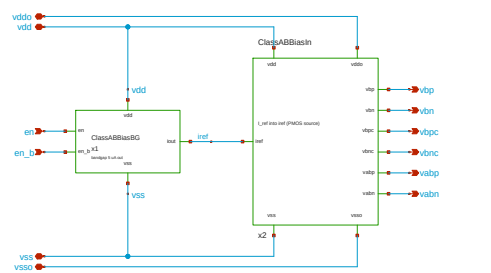
ClassABiasBGdn

Deep-n-well variant: substrate noise to the pair removed, line 0.42 $\rightarrow$ 0.62 %/V, $\approx$ +500–700 μm^2 .

[.sch](#)

ClassABiasBGTree

CACE fixture: ClassABiasBG feeding the ClassABiasIn tree (out $\rightarrow$ ref directly, in the pad BiasRefEnable sits in between). Ports as ClassABiasIn plus en / en_b.



GF180MCU (gf180mcu0), 03v3 devices, sizes from scripbgbg_reference.spice, written by J. J. Schepers, bg.py on 2026-10-09, edit the sheet from now on.

XSCH

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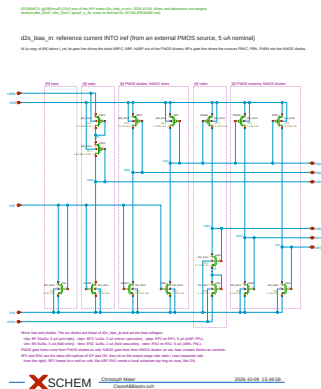
ClassABiasBGTree.sch

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ClassABiasBGTTree

CACE fixture: bandgap core feeding ClassABiasIn.

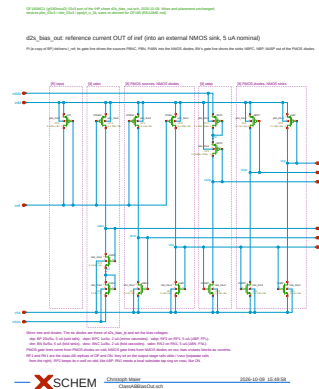
[.sch](#)



ClassABBiasIn

Bias tree and six bias lines, reference current into iref.

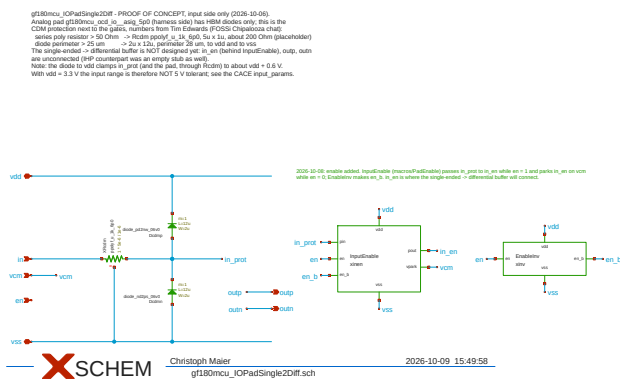
[.sch](#)



ClassABBiasOut

Same tree, reference current out of iref.

[.sch](#)

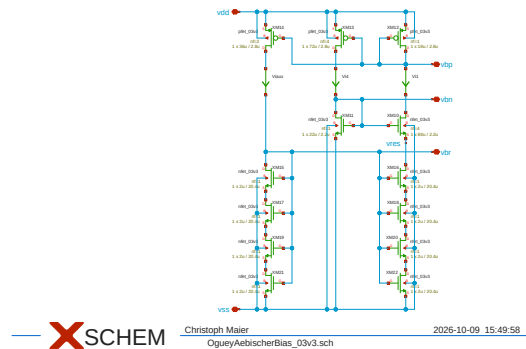


gf180mcu_IOPadSingle2Diff

Input pad: CDM front end (placeholder sizes), InputEnable parks the input on vcm.

[.sch](#) [README](#)

GF180MCU port (03v3 devices) of the IHP sg113cmosf design, 2026-10-06. Wires and placement unchanged, WL re-derived for GF180, see macro README.



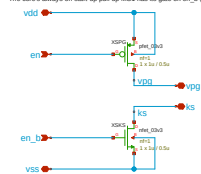
OgueyAebischerBias_03v3

Oguey–Aebischer core, ported wire for wire from IHP, sizes re-derived from gm/ID data. 06v0 variant alongside.

[.sch](#) [06v0.sch](#) [README](#)

RefCoreEnable

Disable switches of a self-biased reference core (ClassABbiasSG), on while en = 0.
 SINC: vpg (PMOS mirror gate line) to vdd → every mirror branch off, out = 0.
 SKS: ks (start-up node) to vss → otherwise ks floats with MS1 off and MS3 would pull vpg.
 The core's always-on start-up pull-up MS1 has its gate on en_b (off while disabled).



GF180MCU (GF180mcu2), GSD3 devices, written by J. Kropf (gm_pndenable.py on 2026-10-09, with the sheet from now on.

XSCH

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RefCoreEnable.sch

2026-10-09 15:49:58

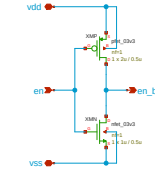
RefCoreEnable

vpg → vdd, ks → vss while disabled.
 Currently unused (switches drawn inline in the BG sheets).

[.sch](#)

EnableInv

en_b = not en, on vdd / vss. One per pad.
 Within the enable context from 1.2 V logic, a level shifter replaces this cell.



GF180MCU (GF180mcu2), GSD3 devices, written by J. Kropf (gm_pndenable.py on 2026-10-09, with the sheet from now on.

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EnableInv.sch

2026-10-09 15:49:58

EnableInv

Local en → en_b inverter.

[.sch](#)

Port documents: [IHP_TO_GF180_PORT_MAP.md](#) · [GF180_PROCESS_OPTIONS.md](#) · [HANDOVER_gf180_migration.md](#)

GF180 · CACE results

Schematic netlists, cloud container. Each cell links to its datasheet (summary.md with plots alongside).

CELL	KEY RESULTS (MIN / TYP / MAX)	FAILS
ClassABDriverBiased	Gain 0.4994 / 0.49999 / 0.50007 · V_{os} 0.013 / 0.021 / 0.719 mV · PM 68.5 / 72.7 / 76.5° · f_c 1.31–1.47 MHz · $I_{Q,OP}$ 205 / 213 / 223 μA · $I_{dd} \leq 307 \mu A$ · settling ≤ 415 ns · noise 249 μV_{rms}	INL max 3.47 mV > 2 mV PM vs load 24.9°
gf180mcu_IOPadDiff2Single	Enabled as driver · disabled I_{dd} 0.078 / 0.093 / 2.9 nA · enable time 1.27–1.52 μs · output glitch at enable 171 / 327 / 489 mV	none glitch accepted for now
ClassABBiasBG	i_{out} error –1.61 / –1.22 / +1.66 % (limit ± 3 %) · mismatch ≤ 4.0 % · line 0.42 %/V max · start-up $\leq 1.3 \mu s$ · I_{dd} 10.0– 10.7 μA · disabled 0.24 nA max	resistor spread –23 / +38 % (no limit)
ClassABBiasBGdn	i_{out} error –1.45 / –0.97 / +1.92 % · line 0.58 %/V max · disabled 1.26 nA max	none
ClassABBiasBGTree / ... dnTree	Six bias lines within –3.0 ... +3.9 % (limit ± 5 %) · I_{dd} 34 μA with tree	none
ClassABBiasIn / ClassABBiasOut	Bias-line errors ≤ 2.9 % / 3.4 % · mismatch ≤ 3.8 % · I_{dd} 24 / 29 μA	none
OgueyAebischerRef_03v3	I_{bias} 90 / 102 / 120 nA · I_q 0.71 μA · noise 0.066 nA/ $\sqrt{Hz}$ · start-up $\leq 16 \mu s$	PSRR 30.7 dB < 50 leg matching 7.5 % > 6
OgueyAebischerRef_06v0	I_{bias} 86 / 107 / 148 nA · start-up ≤ 8.4 μs	I_{bias} max 148 > 130 nA PSRR 35.2 dB leg matching 6.4 %
gf180mcu_IOPadSingle2Diff	Protected gate node –0.30 / 3.45 / 4.30 V	I_{in} 7.5 mA at the top of the input range: not 5 V tolerant with CDM diodes to 3.3 V

Spec limits marked *placeholder* in the yaml files (I_1 70–130 nA, $I_q < 2 \mu\text{A}$, $\pm 15 \%$ / $\pm 6 \%$ mismatch, $I_{in} \pm 1 \mu\text{A}$) are still to be settled.

Flows under development

FLOW	WHAT IT DOES	STATE
Template Makefile flow (both repos)	verify (DRC, LVS, PEX) → build (boundary, LEF, LIB, Verilog stub, GDS, render) → simulate, recursively per macro. Generated outputs committed, so git diff is the regression test.	sg13: proven on template gf180: retargeted, not run
SPICE deck → xschem sheet generators	gen_xschem.py, improvements/xschem/scripts/ (gen_cells, gen_bias, gen_testbenches, export_figures); GF180 gen_bg.py	working
Netlist round-trip checkers	xschem netlist vs reference netlist, device by device: check_xschem.py, check_classab_port.py (now incl. PNPs, 1k resistors, diodes), check_port.py	MISMATCHES 0
CACE characterization	yaml + xschem templates per cell, gen_cace_templates.py , walkthrough cace_chipalooza_walkthrough.pdf	cloud only; IIC-OSIC-TOOLS rerun pending
Clamp PyCells (sg13cmos5l)	Rebuild IHP's flattened ESD clamps as parametrized PyCells; XOR-identical at reference ng. scripts/pcells/README.md	reference cells reproduced
Mixed-signal path	LibreLane harden → magic PEX → XSPICE model in an xschem testbench (generate-xspice), for the scan chain and level shifters	proven with counter
Provenance	Orphan sude1_buecher branch: verbatim chat logs by script (export_chatlog.py), tee'd run logs, MANIFEST.tsv with mtimes	in use

Known traps already documented: full-RC PEX not reproducible (R 574 / 833 / 376 on identical geometry), mismatch MC vacuum on extracted netlists, CACE silently falls back to schematic without -s, xschem < 3.4.8 ignores top_is_subckt, gf180mcuD xschemrc does not set PDK. See [sg13 CLAUDE.md §5](#) and [gf180 handover §7–8](#).

Log passages

GF180 (sudel_buecher)

- [classab_port_log L12-32](#) — task and decisions for the IHP → GF180 driver port
- [L54-78](#) — GF180 sizing of the driver on the ideal fixture; bias variants in / out
- [L105-133](#) — PadEnable cells and pad integration
- [L134-182](#) — CACE: ClassABBiasIn/Out, ClassABDriverBiased, both pads
- [L183-191](#) — xschemrc chain of the whole project (PDK variable trap)
- [L243 ff.](#) — equivalence check of Christoph's hand edits
- [reference_redesign_log L11-22](#) — bandgap task and decisions
- [L38-107](#) — PDK facts checked; core topology and sizing for GF180
- [L108-139](#) — deep-n-well variant; informative 06v0 check at 4.5–5.5 V
- [L140-157](#) — "AVOID POLY RESISTORS!", rescinded; resistor choice
- [L174-214](#) — CACE suites; driver with the real reference
- Run outputs: [logs/main/](#) (CACE .out per cell, sizing tables, [driver_with_bg.txt](#))
- Chat logs: [chatlog/README.md](#) (port survey, bias port + CACE + IOPad PoC, bandgap redesign)

sg13cmos5l (sudel_buecher)

- [improvements/log L22-34](#) — baseline from the first CACE run
- [L74-224](#) — matched-pair DDA, MOS caps, resizing, unit figure of merit, cascoding, noise
- [L484-527](#) — size mockup of d2s_mpdpa
- [L528-607](#) — real bias circuits (in, out, oa, bg) and their sheets
- [L608 ff.](#) — separate output-stage rails vddo / vsso
- [power_down/log L41-94](#) — enable decisions and switch plan
- [L106-119](#) — IHP's clamp practice from sg13cmos5l_io.cdl
- [L120-170](#) — power-down results: leakage, start-up, glitch, unpowered pad
- [class_ab_pad_driver.md](#) — original driver note · [s2d_input_buffer_topology_verbatim.md](#) · [output_driver_sizing](#)
- Chat logs: [chatlog/README.md](#); most relevant: [2026-10-02 class-AB improvements](#), [2026-10-06 power down](#), [2026-09-28 clamp PyCells](#), [2026-09-04 pin mapping and harness review prep](#)

Compiled 2026-10-09 from the four local worktrees and their GitHub remotes. GF180 schematic plates exported with xschem 3.4.8RC and gf180mcu_fd_pr symbols; sg13cmos5l plates are the figures committed on sudel_buecher. Click any plate to open the full-resolution SVG.